Customer Shopping Behavior Analysis

Uncovering spending patterns, payment preferences, and customer segments from **3,900 transactions**

Dataset at a Glance

3,900

Transactions analyzed

18

Feature columns

Demographics

Age, Gender, Location, Subscription

Behavior

Discounts, Ratings, Frequency, Shipping

37 missing values in Review Rating — imputed using median per product category

Made with GAMMA

Data Preparation in Python

01

Load & Explore

Imported with pandas; used `df.info()` and `.describe()`

02

Clean & Standardize

Imputed missing ratings; renamed columns to snake_case

03

Feature Engineering

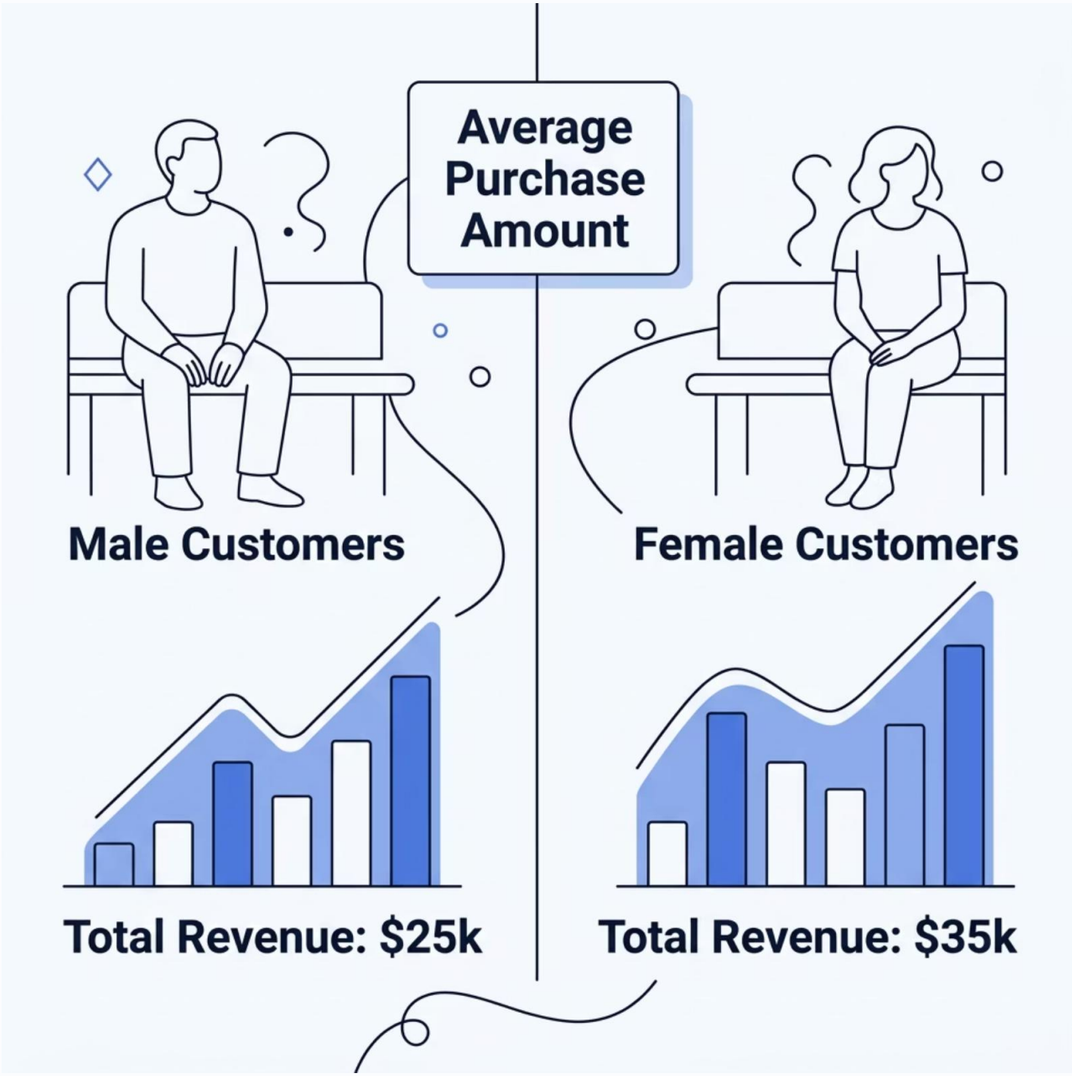
Created `age_group` and `purchase_frequency_days` columns

04

Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis

SQL Analysis: Revenue & Spending



Revenue by Gender

Compared total revenue from male vs. female customers

High-Spend Discount Users

Identified customers using discounts yet spending **above average**

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Products, Ratings & Shipping

☆ Top 5 by Rating

Products with highest average review ratings identified

Shipping Comparison

Average spend: Standard vs. Express shipping

Discount-Dependent

5 products with highest % of discounted purchases

Top 3 per Category

Most purchased products within each category

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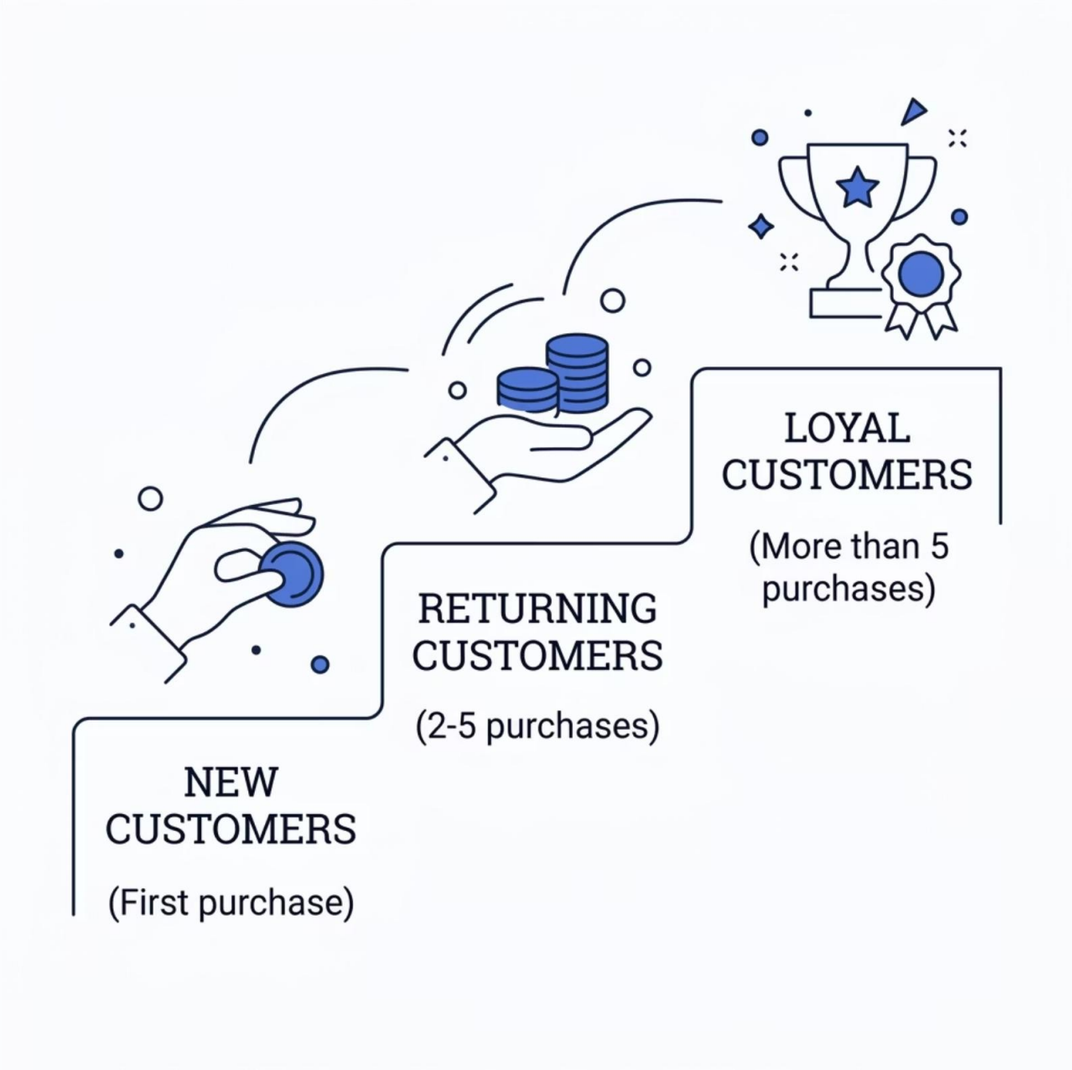
Customer Segmentation



Segment Logic

Categorized by **previous purchase count**

Repeat Buyers & Subscriptions

Customers with **5+ purchases** are more likely to subscribe

Subscribers vs. Non-Subscribers

Compared average spend and total revenue between both groups

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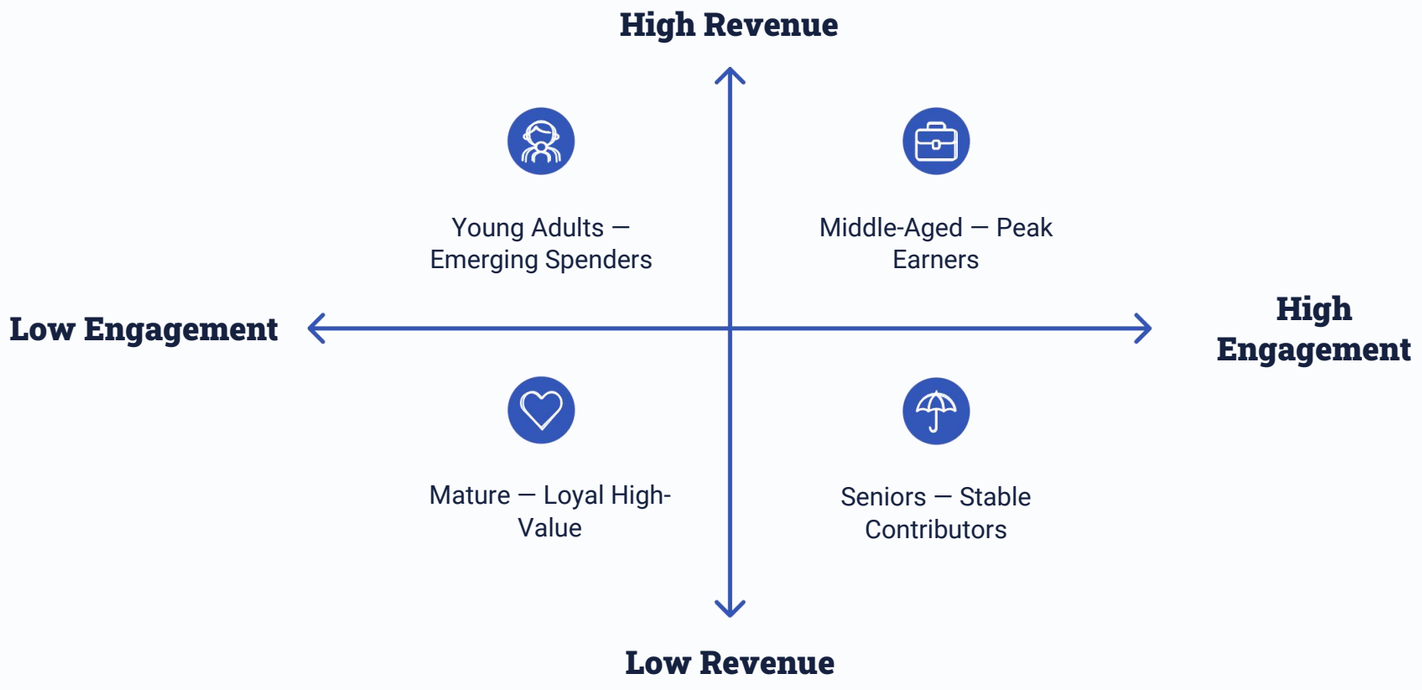
Revenue by Age Group



Age-Driven Insights

Total revenue contribution calculated per age group to identify **high-value demographics** for targeted marketing

Made with GAMMA

Power BI Dashboard

Interactive dashboard built to present all insights visually — enabling stakeholders to explore spending patterns, segments, and product performance at a glance

Made with GAMMA

Customer Behavior Analysis

Subscription Status

No | Yes

Gender

Female | Male

Category

- Accessories
- Clothing
- Footwear
- Outerwear

Shipping Type

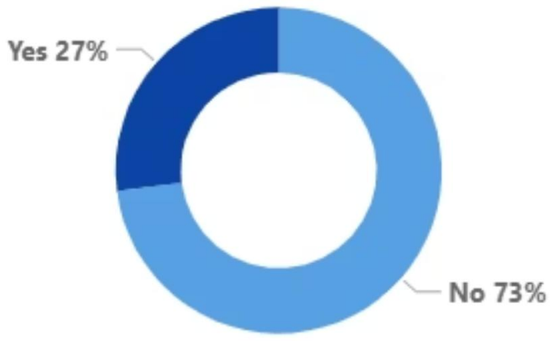
- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

3.9K
Number of Customers

$59.76
Average purchase amount

3.75
Average review rating

% of Customers by Subscription Status



Revenue by Category

Sales by Category

Revenue by Age Group

Sales by Age Group

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to reach "Loyal" tier

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping customers

Product Positioning

Highlight top-rated & best-selling items in campaigns

Made with GAMMA

Key Takeaways

3,900 Transactions

Rich dataset across demographics & channels

Python + SQL

End-to-end pipeline from cleaning to analysis

Actionable Segments

New → Returning → Loyal customer journey

Power BI

Interactive dashboard for strategic decisions